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Role of habitat in determining macroinvertebrate production in an intermittent-stream system

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SUMMARY

1. The effect of channel drying on macroinvertebrate production was studied at the habitat and reach scale in a catchment drained by intermittent streams in Maine, U.S.A. The catchment includes two first-order streams and their second-order confluence. Six reaches were selected for study based on differences in channel slope and habitat cover (bedrock, riffle/run, debris dam and pool). Stream water in each reach was acidic and oligotrophic.
2. The study reaches had different degrees of channel drying. In the first-order reaches, surface flow ceased earlier in the season and for longer periods than second-order reaches. Regardless of reach, pool and debris dam habitats retained water longer than riffle/runs and bedrock. Unlike other habitats, debris dams retained moisture for relatively long periods following cessation of surface flow.
3. Reach-specific macroinvertebrate production ranged from approximately 1.7 to 2.9 g AFDM m^{-2} $year^{-1}$ which are among the lowest values ever reported. Habitat-specific production ranged from approximately 0.5 to 5.0 g AFDM m^{-2} $year^{-1}$ (bedrock and debris dams, respectively).
4. At the reach scale, quantities of stored benthic organic matter (range approximately 200–600 g $AFDM^{-2}$) decreased in a downstream direction.
5. A combination of differences in the timing and duration of channel drying, habitat structure and detritus standing stocks appeared to influence levels of invertebrate production among the study reaches.
6. Our interpretation of a canonical correspondence analysis indicates that drying is more important than habitat in affecting macroinvertebrate production in this intermittent stream system.

Keywords: benthic organic matter, longitudinal gradient, secondary production, stream drying

Introduction

Intermittent streams vary widely in the length and timing of cycles of drying and wetting. This variation has been shown to result in taxon-specific responses that alter invertebrate community structure (Delucchi & Peckarsky, 1989; Dieterich & Anderson, 1995; Erman & Erman, 1995; Jacobi & Cary, 1996). Although some studies have found similar assemblages in intermittent and nearby perennial streams (Delucchi, 1988; Feminella, 1996; Shivoga, 2001), others have shown that invertebrate richness in intermittent streams may be greater or lesser than that of perennial counterparts (Dieterich & Anderson, 2000; Rincón & Cressa, 2000; Meyer, Meyer & Meyer, 2003). Given such effects on community structure, differences in wetting and drying cycles presumably also influence stream productivity.

Numerous studies of perennial streams have shown that habitat heterogeneity affects macroinvertebrate

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community structure (Carter, Fend & Kennelly, 1996; Pardo & Armitage, 1997; Richards *et al*., 1997; Beisel, Usseglio-Polatera & Moreteau, 2000) and secondary production (Smock, Gilinsky & Stoneburner, 1985; Huryn & Wallace, 1987; Wohl, Wallace & Meyer, 1995; Buffagni & Comin, 2000). In intermittent streams, drying regimes probably interact with habitat structure to further affect habitat heterogeneity. For example, pools, bedrock outcrops and debris dams will retain water and moisture for different amounts of time as surface flows cease. As a consequence, differences in habitat heterogeneity may either reduce or magnify the effects of drying because some habitats may provide a refuge (e.g. pools and debris dams) while other may not (e.g. riffle/runs and bedrock outcrops).

In this study we quantified habitat heterogeneity and macroinvertebrate community attributes along an intermittent stream. We predicted that relative production by different taxa, as well as total community production, would: (i) vary systematically along a gradient of stream drying and (ii) vary among habitat types. Further, we evaluated the specific importance of benthic organic matter (BOM) as a determinant of the response of macroinvertebrate productivity to stream drying.

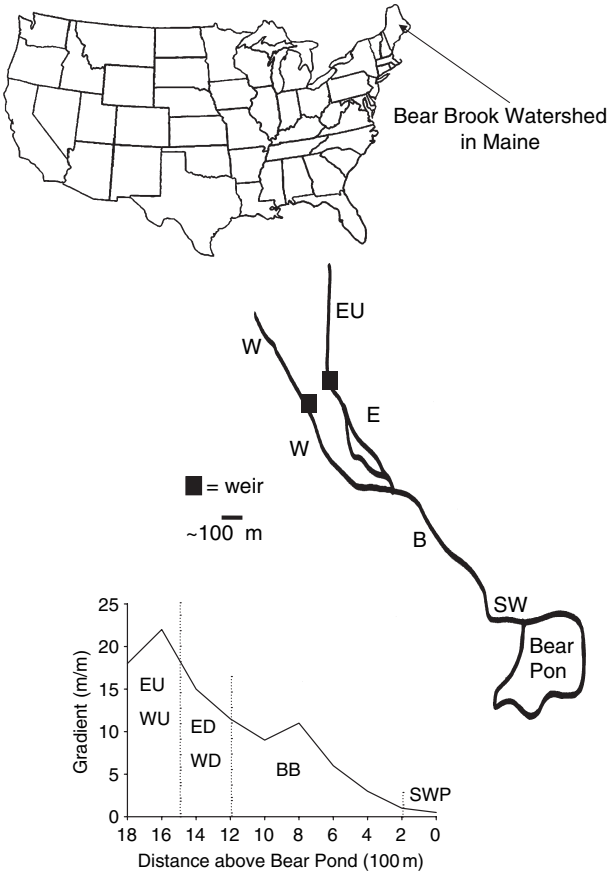

Fig. 1 Map of the location of reaches at Bear Brook in Maine. The inset graph shows channel gradient across all reaches. The reaches include: (i) East Bear Brook above the weir (EU), (ii) West Bear Brook above the weir (WU), (iii) East Bear Brook below the weir (ED), (iv) West Bear Brook below the weir (WD), (v) Bear Brook (upper 800 m of the second order confluence; BB) and (vi) the lower 200 m of Bear Brook which flows through a cedar swamp before entering Bear Pond (SWP).

Methods

Study site

Bear Brook, an intermittent stream, is located on Lead Mountain in the Narraguagus River basin, Maine, U.S.A. Two headwater streams (East and West Bear Brook) and their confluence (Bear Brook) comprise the drainage system (Fig. 1). The upper reaches of East and West Bear Brook are part of a long-term ecosystem experiment investigating the effects of atmospheric deposition on forest ecosystems (Bear Brook Watershed in Maine; Norton & Fernandez, 1999). Both streams have V-notch weirs approximately 300 m above their confluence. Bear Brook flows approximately 1000 m before entering Bear Pond. About 200 m upstream of Bear Pond, the stream enters a cedar swamp where the riparian vegetation changes abruptly from hardwoods [American beech (*Fagus grandifolia* Ehrh.), maple (*Acer* spp.)] to softwoods [Red spruce (*Picea rubens* Sarg.) and Atlantic white cedar (*Chamaecyparis thyroids* L.); Fig. 1].

We divided this stream drainage into six reaches based on stream order, channel slope and habitat attributes (e.g. distribution of bedrock, debris dams, riffle/runs and pools; Table 1). The reaches include: (i) East Bear Brook upstream of the weir (EU), (ii) West Bear Brook upstream of the weir (WU), (iii) East Bear Brook downstream of the weir (ED), (iv) West Bear Brook downstream of the weir (WD), (v) Bear Brook (upper 800 m of the second order confluence; BB) and (vi) the lower 200 m of Bear Brook which flows through a cedar swamp before entering Bear Pond (SWP). Habitat structure (quantified by visual surveys) showed that reaches, excluding SWP, were composed of repeating combinations of riffle/runs, debris-dams and pools. In EU and WU, bedrock-outcrops were also interspersed with the other habitats (Table 1). East Bear Brook (ED) had a braided channel. Unlike the other reaches, SWP contained only pools and runs

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Table 1 Stream order, habitat cover and reach-specific and habitat-weighted channel drying ranks

	EU	WU	ED	WD	BB	SWP
Stream order	1	1	1	1	2	2
Habitat cover (% channel area)						
Bedrock (1)	6	10	0	0	0	0
Riffle/run (2)	48	44	60	75	72	41
Pool (3)	28	25	28	17	19	59
Debris dams (4)	18	21	12	8	9	0
Reach drying rank	1	2	3	4	5	6
Habitat-weighted drying rank	2.78	2.76	2.84	2.91	2.90	2.41

Habitat cover was based on visual surveys conducted when all reaches had flowing surface water. Reach-specific drying ranks are based on observations of the timing of drying and rewetting in each reach. Habitat-weighted drying ranks are the sums of % habitat occurrence × habitat drying rank. Habitat drying ranks are based on visual observation of drying for habitats in each reach and appear in parentheses after each habitat.
EU, East Bear Brook upstream of the weir; WU, West Bear Brook upstream of the weir; ED, East Bear Brook downstream of the weir; WD, West Bear Brook downstream of the weir; BB, Bear Brook (upper 800 m of the second order confluence); SWP, the lower 200 m of Bear Brook which flows through a cedar swamp before entering Bear Pond.

(Table 1). Stream water was acidic (pH range approximately 4.5–6.0) and oligotrophic (NH_4 approximately <20 μg L^{-1}, total phosphorus approximately <8 μg L^{-1}; see Chadwick & Huryn, 2003 for more detail).

Stream channel drying

Stream channel drying was assessed by monthly observations of stream flow in all habitats along all reaches. To accomplish this we walked along the entire stream system and noted where the stream channel was dry. Discharge recorded using the weirs on EU and WU was also used to quantify periods of channel drying for these reaches (Chadwick & Huryn, 2005). Reach and habitat specific drying was quantified using a ranking system based on visual observations of the timing of drying and rewetting in each reach and habitat. Habitat-weighted drying ranks were calculated as the sum of % habitat occurrence multiplied by habitat drying rank.

Macroinvertebrate production

Macroinvertebrates were sampled approximately monthly from November 1998 to May 1999 (i.e. period of surface flow) using a Surber sampler (0.9 m^2, 250 μm mesh). Three samples were taken from each habitat type (e.g. bedrock, debris dams, riffle/runs and pools) in each reach. Sample locations were determined randomly prior to sampling. Samples were preserved in approximately 5% formaldehyde in the field. Samples were later elutriated to remove inorganic particles, and the remaining organic material was sieved into four fractions (>2, 1–2, 0.5–1 and 0.25–0.5 mm). Macroinvertebrates were removed from each fraction under low-power magnification. In some of the benthic samples with large amounts of organic matter, the collected material was subsampled (1/32–1/2) using a Folsom plankton splitter. Once macroinvertebrates were removed, the remaining organic matter was processed further to estimate quantities of BOM (see below). Macroinvertebrates were identified to the lowest taxonomic level possible, assigned to functional feeding groups (Merritt & Cummins, 1996), and body lengths were measured to the nearest 1 mm. Ash-free dry mass (AFDM) was calculated using length–weight relationships (Benke *et al.*, 1999).

Production was estimated using one of three methods. The instantaneous-growth method was used when a cohort could be followed through time and growth rates could be estimated (Benke, 1984, 1993). The size–frequency method was used when a sufficient population size was present (i.e. monthly densities of approximately >5 individuals m^{-2}), but cohorts could not be followed (Benke, 1984, 1993). For this method, cohort production intervals (Benke, 1984) equivalent to the period when flowing water was present were assumed. For rare taxa, production was estimated by multiplying mean annual biomass by an assumed annual biomass turnover rate of 5 (annual production/biomass; P/B; Waters, 1977). Production was estimated for each habitat and then combined to produce a habitat-weighted value for each reach. All estimates of production are expressed as mg AFDM m^{-2} $year^{-1}$.

Benthic organic matter

Benthic organic matter was sorted into various categories (American beech leaves, maple leaves, woody debris, >2 mm miscellaneous and <2 mm miscellaneous), dried at 60 °C to constant mass and then weighed. A portion from each sample was ashed (550 °C) and reweighed to provide information for estimating AFDM. All estimates of BOM are expressed as mg or g AFDM m^{-2}.

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Analysis

BOM and macroinvertebrate production Confidence intervals were calculated for estimates of BOM and secondary production for most macroinvertebrate taxa by bootstrapping (Morin, Mousseau & Roff, 1987; Huryn, 1996). Mean quantities of production and BOM were calculated from vectors of 1000 bootstrap data sets produced by resampling original data sets with replacement (Huryn, 1996). Statistical comparisons between reaches and habitats for both macroinvertebrate production and BOM were made by inspecting the overlap of confidence intervals. Non-overlapping confidence intervals indicated a significant difference. Control of type II error was achieved by establishing an α level of 0.10 prior to beginning the study. We used chi-squared tests to assess differences in the distribution of production among functional-feeding groups and BOM among litter types (Zar, 1984).

Distribution of macroinvertebrate production among habitats and reaches The distribution of macroinvertebrate production among habitats and reaches was evaluated using canonical correspondence analysis (CCA; ter Braak, 1986). CCA is a multivariate technique used to relate community composition to gradients in environmental variables. We conducted CCA using both habitat-specific and habitat-weighted taxon production estimates in the community matrix and BOM, stream order, and ranks of habitat and reach permanence in the environmental matrix. This analysis allowed an assessment of the importance of habitats and reaches in structuring secondary production.

Results

Stream channel drying

Monthly observations indicated complete loss of surface water in each reach from June until autumn. The upper reaches (EU, ED, WU and WD) began drying in April and all were dry by May. Discharge recorded using the weirs showed that EU dried before WU (Table 1, Chadwick & Huryn, 2005). In contrast to the first order reaches, BB and SWP retained isolated pools of water until June. Based upon these observations, we ranked the reaches in order of highest to lowest onset and duration of drying: EU > WU > ED > WD > SWP > BB (Table 1). Regardless of reach, bedrock and riffle/runs dried before pools, and debris dams retained moisture longer than other habitats (Table 1).

Benthic organic matter

Habitat-weighted BOM ranged from 207 to 613 g AFMD m^{-2} and decreased from the upper reaches to BB (Fig. 2). BOM mass in SWP was greater than BB (Fig. 2). With the exception of ED, habitat-specific BOM showed similar patterns across reaches with debris dams > pools = riffle/runs > bedrock (Fig. 2). ED had higher BOM in pools than riffle/runs (Fig. 2). The contribution of different litter types to habitat-weighted BOM was similar among EU, WU, ED, WD and BB ($\chi^2 = 19$, $P = 0.46$). SWP differed from all other reaches, however, because of a low proportion of American beech litter ($\chi^2 = 1.3$, $P = 0.05$; Fig. 3a). The proportional contribution of litter type to habitat-weighted BOM in different reaches differed for debris dams ($\chi^2 = 30.4$, $P = 0.02$), riffle/runs ($\chi^2 = 99.0$, $P < 0.001$) and pools ($\chi^2 = 36.7$, $P = 0.01$), but not bedrock ($\chi^2 =$

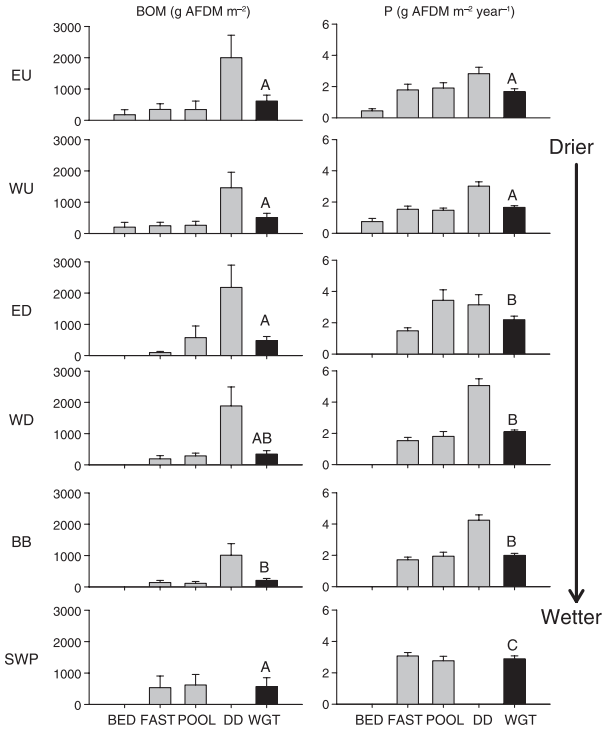

Fig. 2 Habitat-specific and habitat-weighted benthic organic matter standing crop (BOM) and invertebrate secondary production (P) in each reach. Error bars are 90% confidence intervals. Differences among habitat-weighted values are signified by different letter that appear above each bar. WGT, habitat-weighted; DD, debris dams; FAST, riffle/runs; BED, bedrock.

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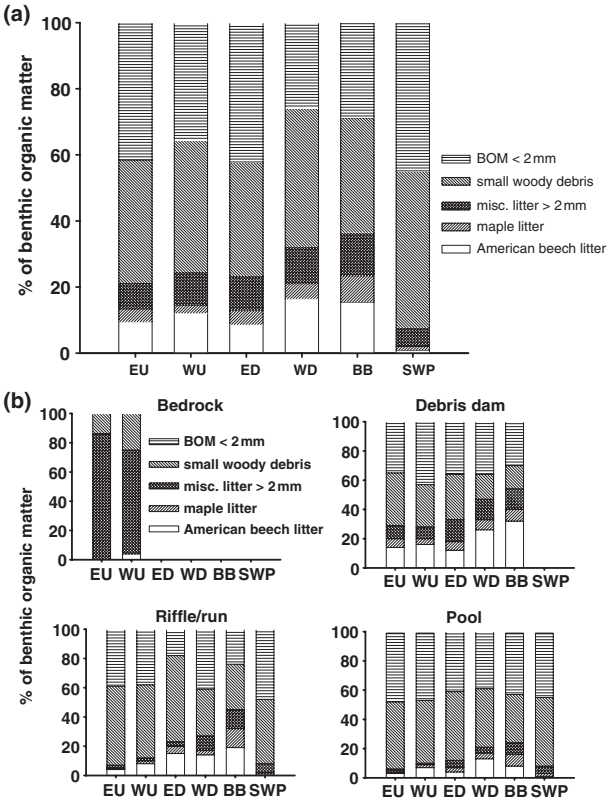

Fig. 3 The distribution of organic matter among benthic organic matter (BOM) types (American beech leaves, maple leaves, small woody debris, >2 mm miscellaneous, or <2 mm miscellaneous) for (a) habitat-weighted (reach-specific) and (b) habitat-specific values.

6.3, $P = 0.18$; Fig. 3b). Riffle/runs and pools in SWP and pools in all other reaches were dominated by BOM < 2 mm and small woody debris. In debris dams and riffle/runs in all reaches except SWP, the proportional contribution of small woody debris decreased and leaf litter increased from up- to downstream. Compared with other habitats, bedrock was dominated by moss (grouped with miscellaneous organic matter).

Macroinvertebrates

Macroinvertebrate richness was similar among reaches (39–49 taxa; Table 2). Habitat-weighted biomass ranged from 0.38 to 0.65 g AFDM m^{-2}, with the lowest occurring in WU and the highest in SWP (Table 2). Habitat-weighted density varied from approximately 3500 to 6000 individuals m^{-2} among reaches (Table 2). Habitat-specific macroinvertebrate richness ranged from 24 to 46 taxa with bedrock having the lowest richness and other habitats having similar values (Table 2). Habitat-specific biomass followed a pattern similar to that of BOM with debris dams > pools = riffle/runs > bedrock (Table 2). Debris dams tended to have the greatest habitat-specific densities while other habitats had similar density. ED had the highest densities of macroinvertebrates in pools rather than debris dams (Table 2).

Total habitat-weighted production increased as the potential for channel-drying decreased (Fig. 2). The lowest production was found in EU and WU (1.68 ± 0.18 and 1.65 ± 0.11 g AFDM m^{-2} $year^{-1}$, ±90% CI), followed by ED, WD, BB (2.19 ± 0.23, 2.09 ± 0.12 and 2.0 ± 0.13) and SWP (2.89 ± 0.19).

Habitat-specific annual production ranged from approximately 0.5 g AFDM m^{-2} in EU bedrock to approximately 5.0 g AFDM m^{-2} in WD debris dams (Fig. 2). Productivity tended to track organic matter standing crop with debris dams > pool > riffle/runs > bedrock. Further, the highest production occurred in reaches that retained channel water the longest (WD, BB and SWP).

Distribution of macroinvertebrate production among habitats and reaches

Canonical axes 1 and 2 accounted for 29% of the variance in the taxon–environment ordination (total inertia = 1.7; Fig. 4). Axis 1 (eigenvalue = 0.30) explained 18% of the variance and separated taxa along a longitudinal gradient (up- to downstream) that also corresponds to stream drying. Axis 2 (eigenvalue = 0.19) explained 11% of the variance and separated taxa based on BOM standing crop and habitat-specific patterns in drying. Our interpretation of the first two canonical axes indicates that drying appears more important than habitat in structuring macroinvertebrate production in this intermittent stream system. Secondary production values used for this analysis appear in Appendix 1.

Discussion

Both organic matter biomass and stream flow regimes are important variables regulating habitat heterogeneity (Frissell *et al.*, 1986; Poff & Ward, 1989; Ward, 1989; Hildrew & Giller, 1994). Numerous studies of perennial streams have shown that habitat heterogeneity affects macroinvertebrate community structure (e.g. Carter *et al.*, 1996; Pardo & Armitage, 1997;

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Table 2 Habitat-weighted (reach-specific) and habitat-specific invertebrate richness, biomass (mg AFDM m^{-2}), and density (individuals m^{-2})

Reach	Habitat	Richness, no. of taxa	Biomass, mg AFDM m^{-2}	Density, ind m^{-2}
EU	Habitat-weighted	40	427 (31)	3570 (221)
	Bedrock	24	88 (23)	1324 (654)
	Riffle/run	31	680 (167)	3191 (809)
	Pool	32	467 (154)	2240 (902)
	Debris dams	36	573 (72)	7082 (2182)
WU	Habitat-weighted	41	382 (25)	3189 (449)
	Bedrock	25	183 (53)	3339 (1192)
	Riffle/run	38	330 (102)	3209 (1582)
	Pool	37	341 (29)	2186 (824)
	Debris dams	36	717 (57)	4040 (1912
ED	Habitat-weighted	41	538 (137)	5870 (1873)
	Riffle/run	39	421 (162)	3338 (851)
	Pool	29	841 (306)	10 418 (5684)
	Debris dams	33	868 (363)	7921 (2414)
WD	Habitat-weighted	39	467 (70)	3482 (556)
	Riffle/run	36	732 (242)	3175 (607)
	Pool	30	377 (91)	2671 (773)
	Debris dams	37	1244 (246)	8074 (1987)
BB	Habitat-weighted	46	456 (100)	2834 (675)
	Riffle/run	38	436 (149)	2702 (852)
	Pool	42	454 (135)	2462 (920)
	Debris dams	42	999 (245)	4677 (1598)
SWP	Habitat-weighted	49	651 (122)	6135 (1390)
	Riffle/run	46	688 (131)	6301 (1746)
	Pool	42	626 (156)	6020 (1922)

Values in parentheses are 90% confidence intervals.
EU, East Bear Brook upstream of the weir; WU, West Bear Brook upstream of the weir; ED, East Bear Brook downstream of the weir; WD, West Bear Brook downstream of the weir; BB, Bear Brook (upper 800 m of the second order confluence); SWP, the lower 200 m of Bear Brook which flows through a cedar swamp before entering Bear Pond; AFDM, ash-free dry mass.

Richards *et al.*, 1997; Beisel *et al.*, 2000) and secondary production (Smock *et al.*, 1985; Huryn & Wallace, 1987; Wohl *et al.*, 1995; Buffagni & Comin, 2000; Boulton, 2003). In the stream reaches that we studied, channel drying and organic matter supply were potentially the most important factors that influenced invertebrate populations and communities. We contend that similar levels of habitat-specific production among reaches were driven by organic matter supply (*sensu* Wallace *et al.*, 1999) but the structure of the communities present in each habitat and reach resulted from differing degrees of channel drying.

Community structure

A number of studies have shown that stream channel drying has a strong effect on macroinvertebrate communities (e.g. Delucchi, 1988; Feminella, 1996; Rincón & Cressa, 2000; Shivoga, 2001). We found similar results using taxon-specific production rather than abundance, the more commonly used metric. Ordination of taxon-based habitat- and reach-specific production showed that invertebrate communities were more similar within reaches than habitats. This suggests that attributes of each reach act to 'filter' invertebrate populations and so control community structure at smaller spatial scales (i.e. habitat) in this intermittent stream system (*sensu* Poff, 1997). Comparable degrees of drying within a reach could well be responsible for similar invertebrate communities that occur among habitats within any given reach. Thus, differences in drying among reaches could be the primary factor that regulates community structure throughout this intermittent stream system.

Secondary production

The amount of organic matter retained within a given habitat was closely related to overall patterns of invertebrate productivity (Fig. 5). A similar positive

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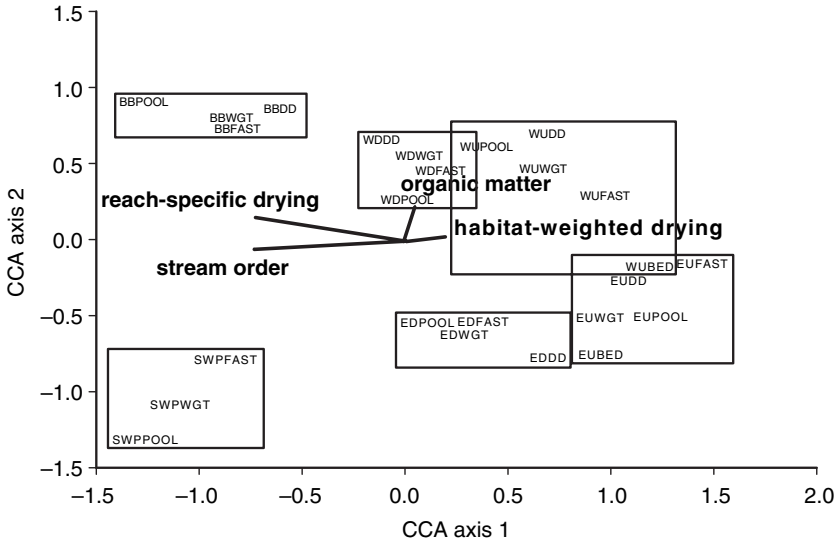

Fig. 4 The taxon–environment ordination from the canonical correspondence analysis showing the locations of habitat assemblages with each reach. Each point is coded by reach [East Bear Brook above the weir (EU), West Bear Brook above the weir (WU), East Bear Brook below the weir (ED), West Bear Brook below the weir (WD), Bear Brook (upper 800 m of the second order confluence; BB) and the lower 200 m of Bear Brook which flows through a cedar swamp before entering Bear Pond (SWP)] and then habitat (WGT, habitat-weighted; DD, debris dams; FAST, riffle/runs; BED, bedrock). Bold lines show the relationship of the environmental variables in the ordination. Boxes have been drawn around points based on reach location. See Appendix 1 for secondary production values for each taxa used in this analysis.

relationship between organic matter and invertebrate productivity has been shown for perennial stream ecosystems (Wallace *et al.*, 1999; Hall, Likens & Malcom, 2001), and our results further support this relationship. For debris dams, however, secondary production did not increase linearly with organic matter, as shown for the remaining habitats in this study (Fig. 5), or in the studies of Wallace *et al.* (1999) and Hall *et al.* (2001). This lack of a linear relationship may be because the material retained by debris dams in the Bear Brook system is often composed primarily of American beech litter, which is unpalatable to macroinvertebrates in Bear Brook (Chadwick & Huryn, 2003). Consequently, a simple increase in organic matter standing crop may not translate directly into higher productivity as seen in other studies (Wallace *et al.*, 1999).

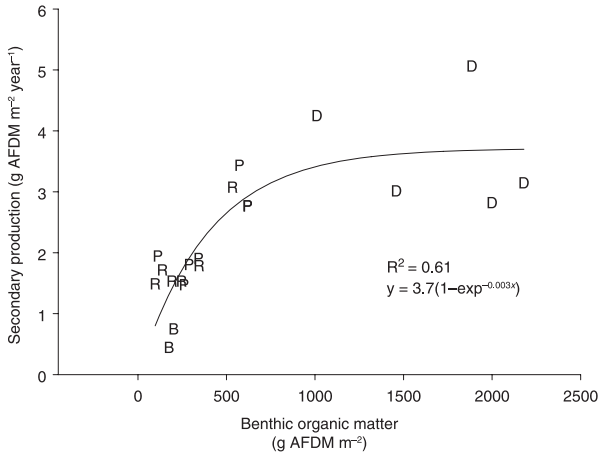

Fig. 5 The exponential rise to max relationship between habitat-specific invertebrate secondary production and organic matter. B, bedrock; D, debris dam; R, riffle/run; P, pool.

Debris dams did, however, provide refuge from drying because moisture was retained for long periods after stream flow ceased. Further, because of their ability to retain water, debris dams possessed the longest and most defined gradient of stream channel drying. As such, macroinvertebrates inhabiting debris dams showed the clearest response to channel drying. As an example, the highest productivity for debris dams occurred in WD and BB (i.e. 'wet' reaches with relatively short drying cycles) when compared with other reaches. The potential for stream drying to be the principal factor driving the differences in secondary production is strengthened given that WD and BB also have low and intermediate habitat-weighted levels of BOM.

Estimates of production for the Bear Brook stream system are among the lowest recorded (Benke, 1993; Huryn & Wallace, 2000) and are plausibly driven by

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stream drying (Chadwick & Huryn, 2005). However, low pH may have also contributed to decreased productivity. For comparison, macroinvertebrate secondary production of two acidic streams were found to be about 2.5 g m^{-2} $year^{-1}$ (Lone Oak Stream; Stead, Schmid-Araya & Hildrew, 2005) and 4.6 g m^{-2} $year^{-1}$ (Broadstone Stream; Woodward, Speirs & Hildrew, 2005). Regardless of the environmental factors that are responsible for regulating productivity, low secondary production is due to reduced macroinvertebrate biomass and/or growth rates (Huryn & Wallace, 2000).

The effect of drying on secondary production is a function of the life-history traits (e.g. developmental time, growth rate, body size) of the taxa that are able persist in intermittent streams. Drying typically results in conditions that exclude large-bodied taxa with long generation times and high biomass. The loss of such taxa would tend to lower community biomass and result in low secondary production (Huryn & Wallace, 2000). In this intermittent stream system, low secondary production is clearly driven by a lack of large-bodied taxa typically found in this region (e.g. Margaritiferidae, Unionoidae, Amphipoda, Aeshnidae, Corduligasteridae, Gomphidae, Pteronarcyidae, Perlidae) that require perennial flow to complete their life cycles (Chadwick & Huryn, 2005).

Such low levels of secondary production among the typical stream habitats recognised in this study (e.g. riffles, debris dam pools) have not been reported from other studies. The production estimated for riffles/runs and pools ranged from approximately 1.5 to 3.0 g AFDM m^{-2} $year^{-1}$. For comparison, two headwater reaches in a North Carolina stream with similar morphology, geology and water chemistry yielded 7–7.5 g AFDM m^{-2} $year^{-1}$ of macroinvertebrate production from all habitats (Wohl *et al.*, 1995). Another headwater stream in North Carolina yielded 7.0–9.3 g AFDM m^{-2} $year^{-1}$ in riffles, pools, and on bedrock outcrops (Huryn & Wallace, 1987). Finally, riffle habitats in a second order stream in New Hampshire yielded approximately 4.0 g AFDM m^{-2} $year^{-1}$ of macroinvertebrate production (Fisher & Likens, 1973; Hall *et al.*, 2001). Estimates of production from comparable streams located throughout the Appalachian Mountains, that are substantially greater than those measured at the Bear Brook system suggest that stream-drying is one major factor explaining the low values of secondary production documented in the Bear Brook drainage.

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Appendix 1 Secondary production values (mg AFDM m^{-2} $year^{-1}$) used in the canonical correspondence analysis. Values appear for each reach (EU, WU, ED, WD, BB, SWP) and for all habitats

Taxon	EUWGT	EUDD	EUFAST	EUPOOL	EUBED
Allocapnia	16.26	0.17	22.31	19.44	1.33
Ameletus	0.00	0.00	0.00	0.00	0.00
Amphinemura	11.42	32.16	6.19	2.83	31.12
Chilopoda	7.74	0.00	16.13	0.00	0.00
Chironomidae	5.01	2.36	0.19	16.06	0.00
Clinocera	21.31	24.14	30.75	2.48	25.16
Dicranota	42.01	37.40	62.77	16.71	7.81
Elmidae	0.08	0.46	0.00	0.00	0.00
Eurylophella	0.22	0.17	0.00	0.66	0.00
Hexatoma	2.59	6.72	2.06	1.16	1.17
Homoplectra	26.63	98.29	3.21	26.44	0.00
Hydaticus	8.24	15.90	11.21	0.00	0.00
Hydatophylax	75.62	38.83	0.00	245.12	0.00
Hydracarina	2.66	6.42	2.27	1.43	0.36
Lepidostoma	28.15	108.87	8.23	16.07	1.84
Leuctra	106.93	316.74	545.00	109.42	2.52
Molophilus	13.06	3.45	19.20	11.45	0.24
Neophylax	0.17	0.00	0.00	0.62	0.00
Oligochaeta	35.37	56.48	35.94	27.07	6.23
Oligostomas	0.18	1.00	0.00	0.00	0.00
Oreogeton	7.73	14.66	6.84	5.50	4.62
Orthocladiinae	119.77	219.62	127.14	55.65	60.53
Ostrocerca prolongata	104.07	181.77	115.21	54.12	15.00
Ostrocerca truncata	5.59	5.49	6.56	4.81	1.72
Palaeagapetus	0.02	0.03	0.00	0.00	0.18
Paranemoura	217.27	485.56	209.41	72.55	150.62
Parapsyche	0.05	0.00	0.09	0.00	0.00
Pedicia	41.38	186.79	16.15	0.00	0.00
Polycentropus	97.88	76.42	0.00	300.43	0.00
Probezzia	2.16	5.72	1.83	0.52	1.71
Prosimulium	130.03	275.75	153.86	8.37	69.99

Appendix 1 (*Continued*)

	WUWGT	WUDD	WUFAST	WUPOOL	EUBED
Allocapnia	2.99	7.24	3.07	0.00	1.16
Ameletus	0.00	0.00	0.00	0.00	0.00
Amphinemura	45.49	80.92	56.44	13.78	2.16
Antocha	0.79	0.00	1.72	0.13	0.00
Chilopoda	23.02	27.78	39.05	0.00	0.00
Chironomidae	0.07	0.11	0.10	0.00	0.00
Clinocera	29.29	41.89	10.23	5.13	147.11
Dicranota	12.35	28.62	11.40	5.21	0.18
Eurylophella	0.05	0.00	0.01	0.17	0.00
Hexatoma	5.01	7.79	5.78	3.33	0.00
Homoplectra	111.02	357.46	60.33	32.53	12.88
Hydaticus	9.03	12.73	0.14	25.16	0.02
Hydatophylax	15.77	19.55	2.31	41.71	2.22
Hydracarina	1.90	2.07	1.10	1.73	5.51
Lepidostoma	26.24	59.33	7.15	42.42	0.32
Leuctra	262.15	414.03	481.05	220.94	68.58
Limnophila	0.77	0.00	0.50	2.22	0.00
Limonia	1.62	0.00	0.00	6.46	0.00
Molophilus	2.32	5.07	1.48	2.15	0.67
Neophylax	0.00	0.00	0.00	0.00	0.00
Oemopteryx	14.37	2.30	0.00	55.56	0.00
Oligochaeta	54.45	86.47	46.03	47.51	41.62
Oreogeton	15.62	13.16	28.18	1.85	0.00
Orthocladiinae	132.54	181.33	144.62	45.49	194.75
Ostrocerca prolongata	64.00	99.09	47.17	65.72	60.07
Ostrocerca truncata	105.68	166.26	117.85	62.67	32.37
Palaeagapetus	0.96	0.00	0.37	1.57	4.07
Paranemoura	48.45	24.56	51.01	56.35	67.59
Parapsyche	54.05	212.64	0.00	12.91	61.71
Pedicia	89.21	272.53	23.96	85.74	0.00
Probezzia	4.80	2.17	6.88	5.25	0.00
Prosimulium	44.59	74.17	54.39	8.05	30.69
Pseudostenophylax	105.34	267.54	105.18	8.30	8.04
Psilotreta	15.89	30.52	17.53	7.07	0.00
Pycnopsyche	136.67	332.51	0.00	267.59	0.00
Rhyacophila	68.91	73.26	91.00	42.80	0.99
Soyedina	13.42	8.14	24.65	3.45	0.00
Sweltsa	11.96	0.52	19.60	12.91	0.00
Tanypodinae	95.68	70.23	51.91	230.42	4.86
Tanytarsini	13.13	5.01	6.28	37.19	0.19
Tipula	5.81	21.95	2.75	0.00	0.00
Turbellaria	0.21	0.00	0.48	0.00	0.00
Wormaldia	8.18	4.58	9.93	11.16	0.62

	EDWGT	EDDD	EDFAST	EDPOOL
Agabus	101.73	155.82	40.55	209.66
Allocapnia	19.58	5.96	0.77	65.74
Ameletus	5.92	0.00	9.86	0.00
Amphinemura	13.68	0.00	22.81	0.00
Chilopoda	3.88	0.00	6.47	0.00
Chironomidae	0.93	5.62	0.43	0.00
Clinocera	6.86	9.82	6.49	6.37

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Appendix 1 (*Continued*)

	EDWGT	EDDD	EDFAST	EDPOOL
Dicranota	40.91	102.96	24.24	50.04
Eurylophella	0.01	0.00	0.01	0.00
Hemerodromia	1.53	1.65	1.52	1.52
Hexatoma	54.37	61.72	16.47	132.46
Homoplectra	13.43	58.07	10.77	0.00
Hydaticus	3.24	3.67	1.06	7.73
Hydatophylax	52.62	16.76	0.00	180.74
Hydracarina	3.07	4.42	1.42	6.02
Lepidostoma	35.16	18.96	16.94	81.15
Leuctra	49.75	99.81	95.75	60.70
Molophilus	4.57	21.32	0.46	6.20
Neophylax	0.82	0.00	1.37	0.00
Oligochaeta	217.09	205.48	131.33	405.84
Oreogeton	30.83	40.40	38.94	9.34
Orthocladiinae	333.21	358.15	153.94	706.69
Ostrocerca prolongata	59.70	73.46	40.29	95.39
Ostrocerca truncata	203.45	550.15	84.34	310.10
Paranemoura	157.37	276.73	178.13	61.72
Parapsyche	34.11	12.12	54.42	0.00
Pedicia	9.13	15.38	1.00	23.86
Polycentropus	0.93	0.00	0.00	3.32
Probezzia	5.63	4.72	0.59	16.80
Prosimulium	142.25	203.17	189.34	15.24
Pseudostenophylax	156.19	575.48	87.16	124.43
Psilotreta	3.38	0.00	5.63	0.00
Pycnopsyche	207.82	69.21	21.91	665.61
Rhyacophila	60.49	72.32	80.75	12.01
Siphlonurus	1.21	0.00	0.38	3.49
Soyedina	4.13	21.77	2.54	0.00
Sweltsa	47.11	9.06	76.70	0.00
Tanypodinae	25.01	10.82	8.08	67.36
Tanytarsini	46.04	38.33	24.85	94.76
Turbellaria	30.83	40.40	38.94	9.34
Wormaldia	7.23	3.72	11.30	0.00

	WDWGT	WDDD	WDFAST	WDPOOL
Agabus	0.00	0.00	0.00	0.00
Allocapnia	9.05	1.83	4.27	33.53
Amphinemura	97.53	219.08	104.39	10.08
Antocha	0.10	1.25	0.00	0.00
Chilopoda	21.36	0.00	28.49	0.00
Chironomidae	0.01	0.07	0.00	0.00
Clinocera	0.92	0.62	1.15	0.00
Dicranota	22.50	146.87	12.00	10.28
Eurylophella	3.38	1.71	3.64	3.04
Hemerodromia	5.33	2.54	6.83	0.00
Hexatoma	8.66	13.69	8.17	8.47
Homoplectra	49.74	136.18	51.53	1.16
Hydaticus	0.95	7.59	0.46	0.00
Hydatophylax	26.38	91.15	14.67	47.55
Hydracarina	0.89	1.52	0.64	1.71
Lepidostoma	13.66	62.78	6.03	24.22
Leuctra	37.97	35.55	45.92	9.47
Molophilus	6.02	27.37	5.11	0.00
Neophylax	0.62	0.00	0.83	0.00
Oligochaeta	409.12	852.07	416.54	167.94

Appendix 1 (*Continued*)

	WDWGT	WDDD	WDFAST	WDPOOL
Oreogeton	5.91	23.13	4.60	3.60
Orthocladiinae	54.89	124.53	54.36	24.45
Ostrocerca prolongata	31.68	96.19	27.19	21.12
Ostrocerca truncata	125.23	225.81	78.21	285.30
Paranemoura	78.15	40.65	79.31	90.69
Parapsyche	168.94	303.46	192.89	0.00
Pedicia	82.04	11.08	106.05	9.51
Polycentropus	20.10	63.84	0.00	88.16
Probezzia	4.12	5.13	4.06	3.90
Prosimulium	94.01	69.28	115.46	11.04
Pseudostenophylax	27.33	139.67	11.02	46.43
Psilotreta	6.51	1.19	0.60	35.12
Pycnopsyche	334.49	1278.21	179.63	573.64
Rhyacophila	134.37	326.39	43.50	20.93
Soyedina	41.86	242.00	14.77	67.21
Sweltsa	8.29	45.39	5.06	5.06
Tanypodinae	103.35	132.19	85.78	167.30
Tanytarsini	7.31	22.49	3.71	16.08
Turbellaria	1.37	9.63	0.39	1.77
Wormaldia	50.29	300.28	31.48	15.66

	BBWGT	BBDD	BBFAST	BBPOOL
Agabus	4.51	33.34	0.00	7.95
Allocapnia	8.03	3.99	6.32	16.42
Ameletus	0.45	1.79	0.00	1.50
Amphinemura	60.93	270.72	50.08	2.67
Attenella	0.46	0.19	0.00	2.34
Chironomidae	0.94	0.14	0.93	1.36
Clinocera	1.10	1.60	0.42	3.46
Dicranota	17.69	34.22	14.50	21.98
Elmidae	4.84	3.53	4.40	7.13
Epeorus	0.16	1.22	0.07	0.00
Eurylophella	6.56	11.10	0.71	26.59
Gomphidae	15.54	0.00	21.57	0.03
Hemerodromia	0.74	3.36	0.59	0.07
Hexatoma	149.30	154.35	159.01	110.12
Hydatophylax	35.13	81.97	24.89	51.75
Hydracarina	0.91	1.31	0.86	0.92
Hydroptila	0.12	1.39	0.00	0.00
Lepidostoma	86.82	190.46	43.90	200.36
Leuctra	45.02	55.87	32.99	34.28
Molophilus	1.09	5.66	0.00	3.04
Neophylax	3.62	15.77	1.61	5.51
Oemopteryx	0.51	0.00	0.71	0.00
Oligochaeta	23.95	13.05	26.00	21.32
Oligostomas	0.08	0.29	0.00	0.29
Oreogeton	20.02	41.13	19.98	10.19
Orthocladiinae	95.08	252.02	83.23	65.96
Ostrocerca truncata	23.04	54.62	23.39	6.74
Palaeagapetus	0.68	1.67	0.74	0.00
Paracapnia	15.93	18.24	18.03	6.87
Paranemoura	34.53	138.94	30.32	1.04
Parapsyche	111.62	427.57	101.27	1.19
Pedicia	45.96	98.04	51.07	1.96
Polycentropus	2.12	0.00	0.00	11.17
Probezzia	10.33	5.48	12.31	5.11

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Appendix 1 (*Continued*)

	BBWGT	BBDD	BBFAST	BBPOOL
Prosimulium	166.19	295.07	190.40	13.38
Pseudostenophylax	12.67	0.00	12.87	17.91
Psilotreta	109.84	8.52	54.86	366.18
Pycnopsyche	389.73	1030.68	285.25	482.05
Rhyacophila	119.86	329.31	104.50	78.84
Siphlonurus	9.97	31.43	0.00	37.60
Soyedina	17.03	124.56	6.86	4.66
Sweltsa	102.30	57.24	104.02	117.13
Tanypodinae	43.04	46.08	28.59	96.38
Tanytarsini	30.14	49.95	27.64	30.25
Tipula	73.60	172.11	63.29	66.01
Wormaldia	94.13	179.28	107.36	3.65

	SWPWGT	SWPFAST	SWPPOOL
Agabus	16.58	40.45	0.00
Allocapnia	0.00	0.00	0.00
Ameletus	27.44	66.44	0.33
Amphinemura	19.66	40.90	4.90
Amphipoda	1.10	0.95	1.20
Chilopoda	16.39	0.00	27.78
Chironomidae	18.68	10.14	24.61
Chrysops	315.03	167.66	417.44
Clinocera	2.22	3.72	1.18
Dicranota	14.37	20.87	9.85
Elmidae	5.82	8.93	3.67
Eurylophella	50.32	26.06	67.17
Hagenius	16.08	0.00	27.25
Hemerodromia	0.94	0.49	1.26
Hexatoma	11.51	8.45	13.63
Hydatophylax	62.21	80.30	49.64
Hydracarina	3.32	2.58	3.83
Hydroporus	5.40	0.00	9.15
Lepidostoma	87.72	75.82	95.99
Leuctra	6.21	5.80	6.50
Lype	6.39	1.92	9.49
Molanna	3.35	3.49	3.25
Molophilus	21.04	14.26	25.75
Neophylax	3.80	3.45	4.03
Oemopteryx	1.15	2.80	0.00
Oligochaeta	224.44	54.84	342.30
Oligostomas	40.81	3.10	67.01
Oreogeton	0.51	1.23	0.00
Orthocladiinae	258.43	177.31	314.80
Ostrocerca prolongata	28.76	64.26	4.09
Ostrocerca truncata	45.83	106.77	3.49
Paraleptophlebia	25.93	47.08	11.23
Paranemoura	17.90	41.23	1.68
Parapsyche	12.09	29.50	0.00
Pedicia	83.00	142.81	41.45
Polycentropus	4.29	5.68	3.32
Probezzia	27.94	12.31	38.80
Prosimulium	120.04	276.69	11.18
Pseudostenophylax	34.00	82.94	0.00
Psilotreta	7.86	19.16	0.00
Pycnopsyche	349.88	486.34	255.05
Rhyacophila	73.81	175.73	2.99

Appendix 1 (*Continued*)

	SWPWGT	SWPFAST	SWPPOOL
Shipsa	64.51	136.18	14.71
Sialis	23.04	13.19	29.89
Siphlonurus	76.51	45.25	98.24
Sphaeriidae	116.91	74.97	146.05
Sweltsa	134.01	182.71	100.29
Tanypodinae	189.32	79.22	265.83
Tanytarsini	204.26	199.83	207.34
Wormaldia	12.48	30.44	0.00

EU, East Bear Brook upstream of the weir; WU, West Bear Brook upstream of the weir; ED, East Bear Brook downstream of the weir; WD, West Bear Brook downstream of the weir; BB, Bear Brook (upper 800 m of the second order confluence); SWP, the lower 200 m of Bear Brook which flows through a cedar swamp before entering Bear Pond; WGT, habitat-weighted; DD, debris dam; FAST, riffle/runs; BED, bedrock.

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